

Decoupling the Attribution Window: Foundation-Model Distillation for Delayed Feedback in Ads Retrieval

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Abstract. Ads retrieval must score on the order of 10^6 candidates within milliseconds, which forces a single small model to serve every advertiser objective at once. Those objectives carry different conversion attribution windows, and a hard attribution label is knowable only after its window closes. Consolidation therefore pins every fast objective to the slowest objective’s clock. We observe that a teacher’s soft label has no attribution window, so supervising the consolidated model with a foundation model decouples the training window from the attribution window entirely. We test the delayed-label half of this claim on a public corpus with 1,600 runs spanning five serving-scale capacities, twenty rolling origins and five strategies. Two ceilings separate the options. Censoring correction is capped by the Fisher information surviving in a censored label, which we derive in closed form from the empirical delay distribution before training anything; it predicts 0.868 for a seven-day window and measurement gives 0.883, 0.874 and 0.849. Distillation is not subject to that cap, beating the infeasible oracle at four of five capacities for a one-day window, and its gain is governed by teacher headroom, vanishing when headroom does. Calibration, which ads auctions price on directly, transfers from teacher to student faithfully and so does its instability: the student inherits the teacher’s per-day calibration error rather than estimating its own.

Keywords: Delayed feedback · Conversion rate prediction · Knowledge distillation · Foundation models · Computational advertising

1 Introduction

An ads retrieval stage has to reduce an inventory of order 10^6 candidates to a shortlist within a latency budget of order 10 ms. That budget is what makes the stage hard, and it has three consequences that together frame this paper.

First, the ranking quantity is a price, not an order. Candidates are ranked by expected value, the product of a paced bid and a predicted conversion rate. An error in the predicted rate is an error in what the advertiser pays, so the estimate must be calibrated and not merely correctly ordered. Metrics that are invariant to monotone rescaling, such as AUC, do not certify this.

Second, advertisers do not share one objective. A production system optimises many down-funnel conversion events, and each carries its own attribution window Δ : an install counted within a day, a purchase within a week. The conventional response is one model per objective. At retrieval QPS that response does not survive the latency budget, so the objectives must be consolidated into a single model with one shared trunk and per-objective heads.

Third, consolidation is obstructed by the label, not by the architecture. Two issues recur. The attribution label $\mathbf{1}[d \leq \Delta]$ is not knowable until Δ has elapsed after the click, so a shared training window must end at $T - \max_k \Delta_k$: consolidating K objectives pins every fast objective to the slowest one’s clock. And down-funnel events are orders of magnitude sparser than upper-funnel ones, so a shared trunk trained on hard labels is dominated by whichever objective is densest. Both are properties of the supervision signal.

This suggests the paradigm shift we study: supervise the consolidated model not with hard attribution labels but with the soft predictions of a large foundation model trained offline. A soft label $\hat{\eta}(x)$ is available at click time, carries no attribution window, and is dense on every row, so in principle it dissolves both obstructions at once. The pattern is deployed. Meta’s GEM is an LLM-scale ads foundation model trained across thousands of GPUs whose knowledge is transferred by distillation, representation learning and parameter sharing into hundreds of latency-sensitive vertical models, lifting conversions by 5% on Instagram and 3% on Facebook Feed [5]; that report also names the failure mode, *stale supervision* when the teacher’s signal lags ground truth, and adds a component realigning teacher outputs to recent data.

This paper isolates the delayed-label half of the claim. We fix one objective and vary its attribution window, the axis consolidation manipulates, across five serving-scale capacities. We do not test multi-objective consolidation or sparsity imbalance; those motivate the design and bound what we conclude (Sect. 7).

Our findings separate two ceilings that the delayed-feedback literature conflates. The first is a property of the data and the second of the budget, and only the second can be bought. Specifically: (i) censoring correction is capped by the Fisher information surviving in a censored label, which we derive in closed form (Proposition 1) and validate to within 0.02 — so correction is already at a ceiling that the delay distribution, not the modeller, fixes; (ii) distillation is subject to no such cap (Proposition 2), beating the infeasible oracle at four of five capacities for a one-day window; (iii) its gain is bought with teacher headroom and vanishes when headroom does, and our teacher is already spent against our largest student, which is the concrete sense in which teacher scale is the operative variable; and (iv) calibration transfers from teacher to student faithfully, instability included, which makes per-period teacher calibration a requirement rather than a refinement wherever predictions are prices.

2 Problem Setting

A click occurs at time t with covariates x . If it converts, the conversion follows after a delay $d > 0$; write $d = \infty$ for a click that never converts. Under attribution window Δ the target is

$$y = \mathbf{1}[d \leq \Delta], \quad \eta(x) = \Pr(d \leq \Delta \mid x), \quad (1)$$

and the served quantity is a price proportional to $\eta(x)$, so the level of $\hat{\eta}$ matters and not only its ordering.

The difficulty is not that y is noisy but that it is not yet determined. At an origin T , the moment a training set is assembled, a row clicked at t has been observable for

$$e(t) = \text{clip}(T - t, 0, \Delta), \quad y_{\text{obs}} = \mathbf{1}[d \leq e(t)] \leq y. \quad (2)$$

If $t \leq T - \Delta$ then $e = \Delta$ and $y_{\text{obs}} = y$ exactly. If $t > T - \Delta$ the row is right-censored: $y_{\text{obs}} = 1$ is conclusive, but $y_{\text{obs}} = 0$ is ambiguous, because the click may still convert inside its remaining window.

2.1 Why Consolidation Couples the Windows

Equation (2) has an immediate consequence for the multi-objective setting that motivates this work. A model head for objective k can be trained on hard labels only where those labels are determined, that is on $[0, T - \Delta_k)$. A consolidated model sharing one trunk and one training stream must therefore draw from

$$W_{\text{hard}} = \bigcap_{k=1}^K [0, T - \Delta_k) = [0, T - \max_k \Delta_k), \quad W_{\text{soft}} = [0, T), \quad (3)$$

where W_{soft} is the window available when the supervision is a teacher’s prediction $\hat{\eta}_k(x)$, which is defined at click time for every k . The statement is elementary, but it is binding: under hard labels the fastest objective inherits the slowest objective’s staleness, and the cost is paid by every head. Under soft labels the intersection disappears. This is the sense in which distillation is not one delayed-feedback remedy among several but the one that removes the coupling rather than compensating for it.

Our experiments fix $K = 1$ and vary Δ , which isolates the per-objective cost of sitting on a longer clock without confounding it with multi-task optimisation. The two windows we use bracket the relevant range: waiting withholds 2.2% of the corpus at $\Delta = 1$ d and 15.5% at $\Delta = 7$ d.

2.2 Five Strategies as Estimators

Let $\mathcal{M} = [0, T - \Delta)$ denote the mature rows and $\mathcal{C} = [T - \Delta, T)$ the censored ones, with $|\mathcal{M}| = m$ and $|\mathcal{C}| = c$. Table 1 states each strategy as a choice of training set

Table 1. The five strategies, with the infeasible Oracle and the distillation control. All are evaluated on the same held-out day against the same target (1), and all serve the same student

	training set	target
Oracle	$\mathcal{M} \cup \mathcal{C}$	true y (infeasible at T)
Wait	$\mathcal{M}$	y
Distil	$\mathcal{M} \cup \mathcal{C}$	teacher’s $\hat{\eta}$; teacher trained on $\mathcal{M}$
Distil-Stale	$\mathcal{M}$	the same teacher’s $\hat{\eta}$ (control)
Correct	$\mathcal{M} \cup \mathcal{C}$	y_{obs} ; loss on $\sigma(z)\hat{F}(e)$, serve $\sigma(z)$
Band	$[0, T - u_k), k = 1 \dots K$	$\mathbf{1}[u_{k-1} < d \leq u_k]$, predictions summed
Reweight	$\mathcal{M}$	y , weighted by $\hat{p}_{\mathcal{C}}(x)/\hat{p}_{\mathcal{M}}(x)$

and target. Every strategy serves the same student, so they are indistinguishable in serving cost and differ only in supervision. The Oracle is infeasible at time T , since it uses labels that do not yet exist; it is the denominator for how much of the penalty a feasible strategy recovers.

Correct is the classical joint conversion-delay likelihood of [1], with a non-parametric $\hat{F}$. Under the conditional-independence assumption

$$\Pr(d \leq e \mid x, d \leq \Delta) = F(e), \quad (4)$$

the observed label has mean $\Pr(y_{\text{obs}} = 1 \mid x, e) = \eta(x)F(e)$, so fitting $\sigma(z(x))\hat{F}(e)$ against y_{obs} under log loss is a proper composite likelihood for η . The correction lives in the loss, and $\sigma(z)$ is served with no post-hoc rescaling.

Distil-Stale is not an ablation but a necessary control. Distil changes two things at once relative to Wait: the student inherits a teacher larger than itself, and it trains on covariates Wait had to discard. Distil-Stale removes only the second. It is the experimental counterpart of the stale supervision reported in deployment [5], and Sect. 5.5 shows the attribution is not the one a reader would guess.

3 Two Ceilings

3.1 What Correction Can Recover

Under (4) a censored observation with window e is Bernoulli with mean $\eta F(e)$. Its Fisher information about η is $I(\eta; e) = (\partial_{\eta} \eta F)^2 / (\eta F(1 - \eta F)) = F / (\eta(1 - \eta F))$, against $I_{\text{full}} = 1 / (\eta(1 - \eta))$ for a mature row, so the relative information of a censored row is

$$\rho(F) = \frac{F(1 - \eta)}{1 - \eta F} \in [0, 1], \quad \rho(1) = 1, \quad \rho(F) \approx F \text{ for small } \eta. \quad (5)$$

Proposition 1. *Assume (4), and that excess risk scales as $1/n$ in the effective sample size. Let $\bar{\rho} = |\mathcal{C}|^{-1} \sum_{i \in \mathcal{C}} \rho(F(e_i))$. Then the fraction of the Oracle-minus-Wait gap recovered by Correct is*

$$R = \frac{1/m - 1/(m + \bar{\rho}c)}{1/m - 1/(m + c)} = \frac{\bar{\rho}(m + c)}{m + \bar{\rho}c}. \quad (6)$$

R contains no fitted quantity: $\bar{\rho}$ follows from the empirical delay distribution and the base rate, and m and c from the calendar. On this corpus the delay distribution is strongly front-loaded, with 57.4% of conversions within seven days landing in the first hour, so censored rows retain most of their information even when their window has barely opened. Over the twenty origins used below, $\bar{\rho} = 0.886$ and $R = 0.888$ for $\Delta = 1$ d, and $\bar{\rho} = 0.847$ and $R = 0.868$ for $\Delta = 7$ d.

3.2 Why Distillation Is Not Subject to It

Proposition 1 prices one strategy, but the Oracle bounds a whole family.

Proposition 2. *Let $\mathcal{F}$ be the student class. Consider any strategy that fits $\mathcal{F}$ to a target measurable with respect to the Oracle’s own training data. Asymptotically its risk is at least the Oracle’s, so its recovery satisfies $R \leq 1$. Correct, Band and Reweight are of this form. Distil is not: its target is produced by a fit in a strictly richer class $\mathcal{G} \supset \mathcal{F}$. As the teacher approaches η , Distil’s limiting risk is the approximation error of $\mathcal{F}$ alone, whereas the Oracle additionally carries an estimation term of order $1/(m + c)$. Hence $R > 1$ is attainable for Distil and not for the others.*

The first half is a data-processing argument: y_{obs} , the band indicators and Reweight’s weights are functions of (x, y, d) observed on a subset of the Oracle’s rows, so each is a coarsening of the Oracle’s signal and no fit of $\mathcal{F}$ to it can do better asymptotically. The second half is the approximation-estimation split. Distillation adds no information; the teacher trains on $\mathcal{M}$, which the Oracle also sees. It substitutes a better *estimator* of η , computed in a class the student could not afford to serve, and hands the student a denoised regression target in place of binary labels.

This is the asymmetry that motivates foundation-model teachers, and it composes with (3). The ceiling in Proposition 1 is set by F , a property of advertisers and users. The ceiling in Proposition 2 is set by $\mathcal{G}$, a property of the offline training budget. Only the second can be purchased, and industrial ads foundation models are a supply of it [5]. Both propositions are asymptotic and ignore optimisation error; Prop. 1 additionally assumes (4).

4 Experimental Design

Protocol. We use rolling origins $T = 40, \dots, 59$, each with training windows ending at T minus that strategy’s own wait, and evaluation on the single day

$[(T+1)d, (T+2)d)$ with one idle day in between, held fixed as the window slides. Label (1) is enforced identically in training and evaluation: a row may enter a training set only if its label is knowable at T . Evaluation days carry 230k to 300k rows, and every evaluation label is fully resolved at every Δ (conversions are logged to day 90.955, clicks stop at 61.001). The origin is the unit of replication and all comparisons are paired on it, because the evaluation positive rate swings from 0.1595 to 0.1966 across these twenty days — an order of magnitude more than any effect we measure.

Tasks. We use $\Delta \in \{1d, 7d\}$ and drop $\Delta = 1h$, whose Oracle gap is -0.00002 ($p = 0.842$): there is no penalty to recover, so every strategy ties with Wait by construction.

Capacity Ladder. The premise is that the student stays inside a serving budget, so the entire comparison is repeated at five capacities $S_0, \dots, S_4$. All five share one backbone and differ only in the number of categorical hash-table rows (36, 64, 160, 896 and 131,072, or 2,592 to 9,437,184 table parameters, a $3,641\times$ range), a single monotone knob that is exactly the embedding-table memory footprint and therefore the binding constraint in production retrieval. The ladder was constructed to be approximately equally spaced in normalised entropy (mean step 0.0149, from 0.78626 at S_0 to 0.72794 at S_4 at one reference origin) and is monotone in it at 40 of 40 (task, origin) cells.

Teacher. Distil always uses the same teacher, selected as the best of a 21-rung architecture ladder and held at full table resolution: 57.3M parameters and 1.59M FLOPs per example, against 704 FLOPs for every student, a ratio of $2,262\times$. It is a training-only asset and could not be served under the student’s budget. It sees Wait’s window and Wait’s labels, so it differs from Wait on capacity alone. We define its *headroom* as $H = \text{NE}(\text{Wait}) - \text{NE}(\text{teacher})$.

Metrics. We report normalised entropy $\text{NE} = \text{logloss}/H(\bar{y})$ and the calibration ratio $\bar{p}/\bar{y}$. Because an auction prices on the day it runs, we summarise calibration *per origin* and then average, rather than averaging the ratio first: the two differ sharply (Sect. 5.6). All models train for exactly one epoch; a second epoch already overfits at every capacity (train -0.0186 , evaluation $+0.0019$, $p = 0.025$).

Scale and Verification. The grid is 2 tasks $\times$ 5 capacities $\times$ 20 origins $\times$ 8 arms, or 1,600 scored runs. Gates checked before launch: the harness reproduces previously logged cells bit-exactly (14 of 14); the shared teacher is identical across capacities to seven decimal places while students are not; and every arm at a given (task, origin) is scored on an identical evaluation set.

5 Results

5.1 The Cost of the Clock

The Oracle-minus-Wait gap at $\Delta = 7d$ is $+0.00538$, $+0.00602$, $+0.00746$, $+0.00842$ and $+0.00598$ for S_0 to S_4 , all with $p < 0.001$ over 20 of 20 origins. At $\Delta = 1d$

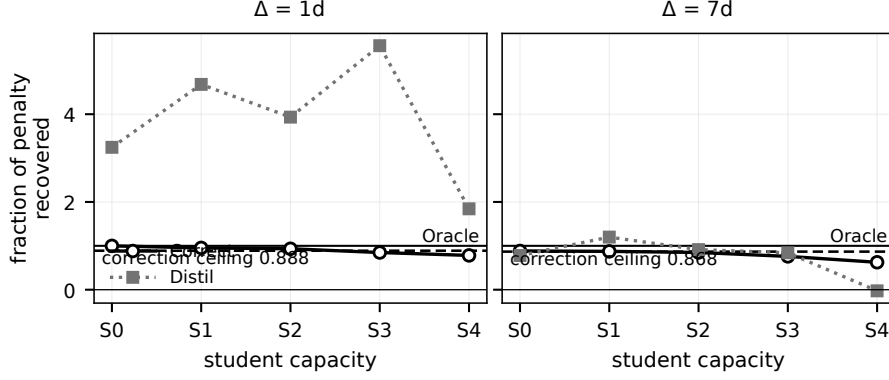


Fig. 1. The two ceilings. Correct (open circles) tracks the parameter-free prediction of Proposition 1 (dashed) at the three smallest capacities and never crosses the Oracle line at 1.0. Distil (grey squares) is governed by neither bound: it exceeds the Oracle throughout $\Delta=1d$ and collapses at S_4 for $\Delta=7d$, where the teacher has almost no headroom left

it is an order of magnitude smaller, from $+0.00067$ to $+0.00092$, though equally consistent (20 of 20, $p < 0.001$). Moving a head from a one-day to a seven-day clock therefore costs roughly $8\times$ more, which by (3) is what consolidation imposes on a fast objective. Because the $\Delta = 1d$ denominator is below 0.001 NE, we report absolute differences at that window and treat recovery *ratios* there as indicative only.

5.2 Correction Attains Its Ceiling, and Then Falls Below It

For $\Delta = 7d$ the predicted ceiling is 0.868, and Correct measures 0.883, 0.874, 0.849, 0.759 and 0.625 across S_0 to S_4 . At the three smallest capacities the prediction is accurate to $+0.015$, $+0.006$ and -0.019 , from a formula with no free parameters fitted to nothing. At S_3 and S_4 it over-predicts by 0.11 and 0.24.

The shortfall is misspecification, not estimation. Assumption (4) is false here in a way we can measure: stratifying converted rows by one categorical field gives six strata of 23,790 to 860,851 conversions, across which the first-hour share of within-seven-day conversions ranges from 0.453 to 0.689 (pooled 0.574) and the median delay from 0.16h to 1.73h. A model too small to resolve covariate-dependent delay cannot be harmed by a correction assuming it away, so at S_0 to S_2 the only cost of censoring is the statistical one Proposition 1 prices; as capacity grows the bias in $\hat{F}(e)$ becomes expressible and recovery falls below the ceiling. Either way the practical reading is the same: where the ceiling is attained nothing is left for a better correction to win, and where it is not, the remedy is a covariate-conditional delay model rather than a tighter estimator.

5.3 Distillation Is Not Bounded by the Oracle

No correction-based strategy exceeds the Oracle at $\Delta = 7$ d: Correct is below it at every capacity by 0.00063 to 0.00224 ($p < 0.005$), Band by 0.0035 to 0.0082, and Reweight by 0.0039 to 0.0081, all $p < 0.001$. This is Proposition 2’s first half, and Correct’s placement just under the bound is consistent with it being an efficient rather than a wasteful estimator.

Distil behaves differently. At $\Delta = 1$ d it beats the infeasible Oracle at S_0 to S_3 by +0.00150, +0.00294, +0.00269 and +0.00397, all $p < 0.001$, and by +0.00071 ($p = 0.06$) at S_4 . At $\Delta = 7$ d it is statistically indistinguishable from the Oracle at S_0 to S_3 ($p = 0.45$ to 0.73) while every other strategy is significantly below it. Against Wait it gains +0.00217 to +0.00484 at $\Delta = 1$ d and +0.00419 to +0.00722 at $\Delta = 7$ d over S_0 to S_3 , all $p < 0.001$ at 19 or 20 of 20 origins. Head to head with Correct it wins outright at the short window, by +0.00150 to +0.00411 at S_0 to S_3 ($p < 0.001$, 20 of 20 origins) and +0.00089 at S_4 ($p = 0.019$), and ties at the long one (± 0.0020 , $p = 0.25$ to 0.80). The asymmetry follows from Sect. 5.2: at seven days Correct is near a ceiling worth 0.868 of a large penalty, whereas at one day the penalty is small and the teacher’s advantage is not.

One result cuts against Proposition 2 and we report rather than reconcile it. At $\Delta = 1$ d, Reweight also exceeds the Oracle at S_0 (+0.00199) and S_2 (+0.00041), $p < 0.001$. The proposition is asymptotic, and Reweight trains on strictly less data under non-uniform weights, which regularises; at a window whose entire penalty is under 0.001 NE we read these as finite-sample effects. They do not recur at $\Delta = 7$ d, where Reweight is below the Oracle at all five capacities.

5.4 The Gain Is Bought With Teacher Headroom

Teacher headroom H falls monotonically as the student grows, from +0.0572 at S_0 to +0.0036 at S_4 for $\Delta = 7$ d, because the teacher is fixed while the student closes on it. Distil’s gain does not follow H down. Across S_1 to S_3 , where H falls $2.8\times$ from 0.0484 to 0.0172, the gain is essentially flat at +0.00722, +0.00682 and +0.00709: a larger student extracts a larger fraction of a smaller advantage, with transfer efficiency gain/ H rising from 14.9 % to 41.2 % (and 7.6 % to 31.0 % at $\Delta = 1$ d).

The flatness ends abruptly. At S_4 , $\Delta = 7$ d, headroom is +0.0036 and Distil collapses to -0.00015 ($p = 0.898$) — the only non-positive cell in the grid, and the only one where the Distil-Stale control is itself negative (-0.00126), meaning the teacher is actively degrading the target. Distillation helps exactly while the teacher is meaningfully better than the student, which is a statement about model size rather than about feedback delay. Our teacher, at $2,262\times$ the student’s inference cost, is already spent against the largest student we tested; scaling the teacher is therefore not an optional refinement of this result but the direction in which it continues.

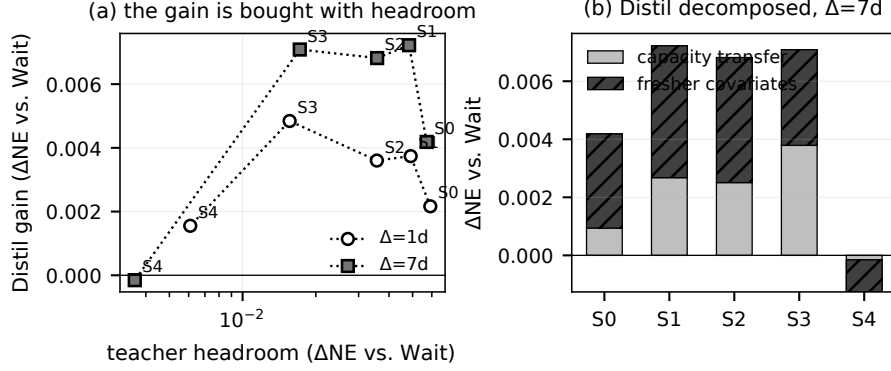


Fig. 2. (a) Distil’s gain against teacher headroom H (log axis), one point per capacity and window. Gain is roughly flat while H falls by $2.8\times$, then collapses at the single cell where H reaches $+0.0036$. (b) Distil decomposed with the Distil-Stale control at $\Delta=7d$. The fresher-covariate term is 47 to 78 % of the gain

5.5 What Distillation Is Actually Buying

Read alone, Distil’s win over Wait looks like delay mitigation. The Distil-Stale control shows it is only partly that. Capacity transfer accounts for 22 to 53 % of the gain at $\Delta = 7d$ across S_0 to S_3 , and 71 to 94 % at $\Delta = 1d$ (Fig. 2b). The residual, the student’s exposure to covariates Wait discarded and which carry no new label information, is what is attributable to the delayed label; it is significant at every capacity ($p \leq 0.0013$, 19 or 20 of 20 origins) and worth $+0.0033$ to $+0.0045$ at $\Delta = 7d$. Reporting Distil against Wait alone would overstate delay mitigation by a factor of two to four at $\Delta = 1d$, and we would encourage this literature to treat a Distil-Stale-style control as mandatory.

Band and Reweight do not repay their complexity: Band spans -0.00215 to $+0.00457$ with no pattern in capacity and is significant in 5 of 10 cells, its slowest band still training on Wait’s window; Reweight stays within -0.00055 to $+0.00265$ of Wait everywhere.

5.6 Calibration Transfers, Including Its Instability

Because the served quantity is a price, calibration is not a diagnostic here but part of the objective. Table 2 reports it two ways, and the two disagree.

On long-run bias Distil looks best of the feasible strategies and the teacher best of all. Per origin the ranking inverts: Distil is the worst-calibrated arm, exceeding Wait’s daily error by 0.019 ($p = 5.7 \times 10^{-4}$, paired over 100 (capacity, origin) cells) and the Oracle’s by 0.020 . The mechanism is visible in the per-capacity values, which we give because they are unusually clean: the teacher’s daily error is 0.0501 at every capacity by construction, and Distil’s is 0.0597 , 0.0508 , 0.0498 , 0.0519 and 0.0501 at S_0 to S_4 , while Wait and the Oracle sit

Table 2. Calibration at $\Delta=7$ d, averaged over the five capacities. Averaging the ratio over origins first lets opposite-sign daily errors cancel; an auction prices on the day it runs, so the middle column is the operative one

	long-run bias	error on a day-to-day	
	$ \bar{c} - 1 $	serving day $ \bar{c} - 1 $	swing (sd)
Oracle	0.0153	0.0320	0.0381
Wait	0.0113	0.0332	0.0400
Correct	0.0124	0.0339	0.0410
Reweight	0.0116	0.0403	0.0497
Teacher	0.0025	0.0501	0.0697
Distil	0.0100	0.0524	0.0691

at 0.028 to 0.041. The student does not estimate its calibration from labels; it copies the teacher’s, and it copies the day-to-day variance with it.

Calibration is thus a transferable property: a well-calibrated foundation model confers calibration on every student it teaches at no serving cost, a second axis along which teacher quality pays. But transfer is faithful in both directions, so where predictions are prices the teacher must be calibrated *per serving period*, not merely unbiased in the long run — accuracy alone is not a sufficient teacher-selection criterion. That this is operational rather than an artefact of our setup is supported independently: the deployed system we take as motivation reports stale supervision from teacher lag and adds a component realigning teacher outputs to recent ground truth [5].

6 Related Work

[To be written. Anchor on the joint conversion-delay likelihood [1]; the importance-weighting line [4, 8, 6]; stream-level alternatives [3, 2]; covariate-conditional delay [7, 9], which targets exactly the assumption Sect. 5.2 shows is violated; and industrial ads foundation models with post-training knowledge transfer [5]. Distillation and multi-task-consolidation references still to be added, and verified.]

7 Limitations and Outlook

The scope is deliberate and it bounds the conclusions. We motivate the work with multi-objective consolidation but test $K = 1$: (3) is an argument, not a measurement, and we do not show that a consolidated multi-head model behaves as the union of our single-objective cells. Sparsity imbalance across objectives, the second obstruction we cite, cannot be studied on a corpus with one conversion event, so it appears here only as motivation. And the extrapolation to LLM-scale teachers is untested: we show the gain is governed by teacher headroom and that our teacher’s headroom is exhausted at S_4 , not what three further orders of magnitude of teacher would deliver. What the grid establishes is that

the mechanism is real, that it is not bounded by the quantity bounding every alternative here, that the alternative is already at its own ceiling, and that calibration transfers along with accuracy.

On the measurements: one seed per origin, with the origin as the unit of replication ($n = 20$); seed sensitivity was characterised separately at a standard deviation of 0.0004 to 0.0030, an order of magnitude below the $\Delta = 7$ d effects, but seeds are not averaged and the origins share training data heavily, so the effective sample size is below twenty. Proposition 1 assumes a $1/n$ excess-risk rate we do not verify directly; Proposition 2 is asymptotic, with a finite-sample exception in Sect. 5.3. Distil’s soft targets are largely teacher in-sample on $\mathcal{M}$, and out-of-fold targets are untested. All results come from one public corpus with one delay distribution, so Proposition 1’s ceiling is corpus-specific even though its formula is not.

Acknowledgments. *[To be added after review.]*

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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